

Assignment 1

1. For which values of a will the following system have no solutions? Exactly one solution? Infinitely many solutions?

$$\begin{aligned} x + 2y - 3z &= 4 \\ 3x - y + 5z &= 2 \\ 4x + y + (a^2 - 14)z &= a + 2 \end{aligned}$$

2. Let $A = [a_{ij}]$ be an $n \times n$ matrix. Determine whether A is symmetric.
 - a. $a_{ij} = 2 + j^2$
 - b. $a_{ij} = 2 - j^2$
3. Let A be a square matrix such that $A^4 = 0$, show that.

$$(I - A)^{-1} = I + A + A^2 + A^3$$

Assignment 2

Linear Algebra

1. Find the coordinate vector of A relative to the basis $S = \{A_1, A_2, A_3, A_4\}$. Where,

$$A = \begin{pmatrix} 2 & 0 \\ -1 & 3 \end{pmatrix}, \quad A_1 = \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad A_4 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

2. If $W = \{(a, b, c, d) \mid a, b, c, d \in \mathbb{R} \text{ where } d = a + b, \quad c = a - b\}$,
- Show that W is a vector subspace of $\mathbb{R}^4$.
 - Find a basis for W .
 - Find the dimension of W .